

① Question

- i (
- ii (
- iii) $\rightarrow$ balance parentheses
- iv [
- v {
- vi)
- vii {
- viii [
- ix]
- x }
- xi)
- xli)

{ [] }

6 token S

(, { [] }

② $i=1$ $j=2$ $1/2 \neq 0$ $\rightarrow$ push(1) [1]
 $j=3$ $1/3 \neq 0$ $\rightarrow$ push(1) [1,1]
 $j=4$ $1/4 \neq 0$ $\rightarrow$ push(1) [1,1,1]

$i=2$ $j=2$ $2/2 = 0$ $\rightarrow$ pop [1,1]
 $j=3$ $2/3 \neq 0$ $\rightarrow$ push(2) [1,1,2]
 $j=4$ $2/4 \neq 0$ $\rightarrow$ push(2) [1,1,2,2]

$i=3$ $j=2$ $3/2 \neq 0$ $\rightarrow$ push(3) [1,1,2,2,3]
 $j=3$ $3/3 = 0$ $\rightarrow$ pop(3) [1,1,2,2]
 $j=4$ $3/4 \neq 0$ $\rightarrow$ push(3) [1,1,2,2,3]

$i=1, j=1, 1 \times 1 = 0$ pop $[1, 1, 2, 2, 3]$

$i=4, j=2, 4 \times 2 = 0$ pop $[1, 1, 2, 2,]$

$j=3, 4 \times 3 \neq 0$ push(4) $[1, 1, 2, 2, 4]$

$j=4, 4 \times 4 = 0$ pop $[1, 1, 2, 2]$

$i=5, j=2, 5 \times 2 \neq 0$ push(5) $[1, 1, 2, 2, 5]$

$j=3, 5 \times 3 \neq 0$ push(5)

$j=4, 5 \times 4 \neq 0$ push(5) $[1, 1, 2, 2, 5, 5, 5]$

$i=6, j=2, 6 \times 2 = 0$ pop

$j=3, 6 \times 3 = 0$ pop

$j=4, 6 \times 4 \neq 0$ push(6) $[1, 1, 2, 2, 5, 6]$

$i=7, j=2, 7 \times 2 \neq 0$ push(7)

$j=3, 7 \times 3 \neq 0$ push(7)

$j=4, 7 \times 4 \neq 0$ push(7) $[1, 1, 2, 2, 5, 6, 7, 7, 7]$

$i=8, j=2, 8 \times 2 = 0$ pop

$j=3, 8 \times 3 \neq 0$ push(8) $[1, 1, 2, 2, 5, 6, 7, 7, 8]$

$j=4, 8 \times 4 = 0$ pop $[1, 1, 2, 2, 5, 6, 7, 7]$

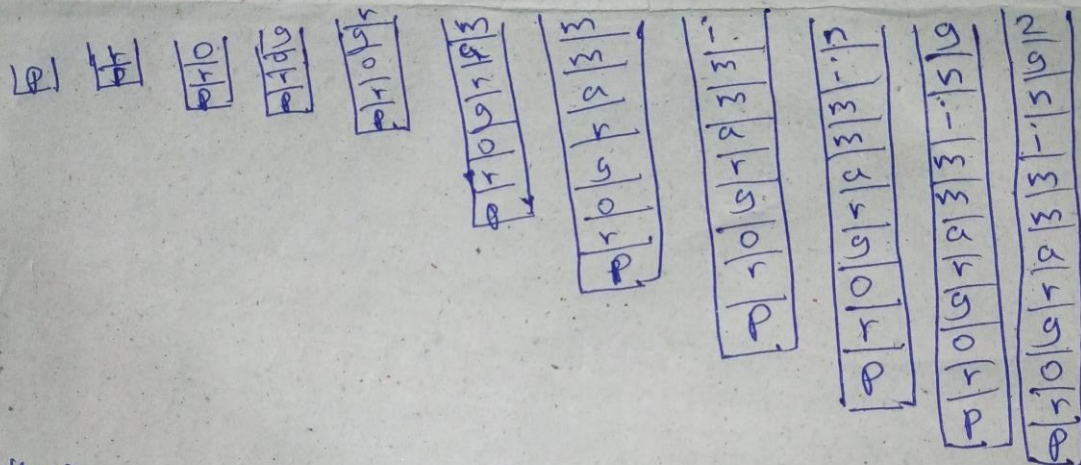
$i=9, j=2, 9 \times 2 \neq 0$ push(9)

$j=3, 9 \times 3 = 0$ pop

$j=4, 9 \times 4 \neq 0$ push(9) $[1, 1, 2, 2, 5, 6, 7, 7, 9]$

final contents = $[1, 1, 2, 2, 5, 6, 7, 7, 9]$

③



Stack →

final → 29nimmargorP

④ The $n/2$ operation gets the least significant bit 0 or 1 of and push it onto the stack.

Then $n = n/2$ operation shifts to the right by one bit.

While (!isEmpty(cs)) loop pops and prints each element from the stack until the stack is empty.

(5)

push

```
#define n 4;
int stack[n];
int top = -1;

void push(int data) {
    int data;
    if (top == n-1) {
        printf("overflow");
    }
    else {
        top = top + 1;
        stack[top] = data;
    }
} return 0;
```

pop

```
#include <stdio.h>
#define n n 4;

int stack[n];
int top = -1;

int pop() {
    int data;
    if (top == -1) {
        printf("underflow");
        exit(1);
    }
    data = stack[top];
    top = top - 1;
    return value;
}
```

Empty

```
void is Empty() {  
    if (top == -1) {  
        printf("Stack is Empty");  
    }  
    else {  
        printf("Stack is not empty");  
    }  
}
```

Is full

```
void is full() {  
    if (top == N-1) {  
        printf("Stack is full");  
    }  
    else {  
        printf("Stack is empty");  
    }  
}
```


question no 6 :

$$623 + -382 / + * 2^3 +$$

$$65 - 382 / + * 2^3 +$$

$$1 - 382 / + * 2^3 +$$

$$134 \cancel{2} + * 2^3 +$$

$$17 * 2^3 +$$

$$72^3 +$$

$$49 \mp 3$$

$$\underline{\underline{52}}$$